Finite Scientific Convergence Report: From Anisotropic Observables to Model-Independent Diagnostics and Geometry Compatibility Atlas

HTT / MIO Collaboration

Task ID: `PR-R37-05-FINAL-REPORT` · Claim Tier: `C0_DIAGNOSTIC_ONLY`

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Abstract

We present the final integrated report of the `htt_base` scientific convergence campaign (`CONVERGENCE_20261028`, Units `R37-00` through `R37-05`), establishing a claim-tiered, manifest-backed, and transfer-provenance-aware observational metrology before the external native low-ℓ Bianchi Boltzmann solver arrives. We synthesize five key physical contributions: (1) exact information equivalence between common-carrier representations and joint generalized least squares (GLS) without premature scalar reduction; (2) proof of the linear polarisation omission bias bound ($\|\Delta\boldsymbol{\Sigma}\| \leq 0.0087824$, observed 0.0085038) and a deterministic positive-fibre witness demonstrating that Thomson intensity gain identifies a convex set rather than a unique shear ($\|\Delta\boldsymbol{\Sigma}\| = 1.18 \times 10^{-4}$); (3) Gaia DR3 astrometric depth tomography via GLS sandwich covariance repair (0.94927 empirical coverage) and the exact fixed-Q sign-stability theorem ($\rho_{\text{sign}} = 1/\sqrt{3} \approx 0.57735$); (4) Pantheon+ SNe Ia directional Hubble ratio bounds ($[0.9222, 1.1074]$ at 95% confidence), quantum channel transmission limits ($D_L \in [1.58114, 3.0]$), and non-perturbative Clarkson–Bassett–Lu (CBL) opacity control; and (5) spatial-Ricci orbit separation for Bianchi II and VII_0 witnesses ($\delta(a) = \sqrt{2}a^2u_*^2$) with an exclusive stability margin ($M_{\text{excl}} = +0.36258$). We anchor these methods with the full qualification of Planck PR4 Commander low-ℓ polarisation morphology over 100 end-to-end processed Monte Carlo simulations, proving complete consistency with the standard isotropic null ($p(J_{234}) = 0.4356$, $p(|J_{234}|) = 0.7426$, $p(\text{align}_{23}^2) = 0.2277$). All 208 executable DAG nodes are completed, zero unblocked tasks remain, and the 65 residual tasks held by external solver or mock dependencies are cataloged in an exhaustive audit appendix.

1 Introduction and Scientific Boundaries

Astrophysical searches for large-angle anomalies, anisotropic expansion, and beyond-FLRW geometry frequently struggle with premature scalar projections, conflation of observational features with cosmological truth claims, and uncalibrated Boltzmann solver approximations. The mission of `htt_base` is to construct a rigorous, claim-tiered metrology that separates:

- **Model-Independent Observatory (MIO)**: Diagnostic certificates, depth/directional coherence, and metric spaces without cosmological model assumptions;
- **Hypothesis Testing and Tomography (HTT)**: Model-dependent likelihoods, posterior bundles, and mixture inference;
- **Observable Statistics (`obsstat`)**: Harmonic features, BiPoSH spectra, and null ensembles without inference claims;
- **BASS Pre-Solver Interface**: Strict transfer provenance and adapter schemas awaiting the native 1+3 PSTF Bianchi Boltzmann solver.

Mandatory Non-Overclaim Firewalls (AGENTS.md): No native Boltzmann solver has been simulated. All empirical outputs remain classified as Claim Tier C_0 (diagnostic-only / transfer-conditional). Model-independent diagnostics are strictly barred from generating posterior odds or truth certificates. Low-ℓ morphology features do not identify a Bianchi family.

2 Five-Coordinate Metric Governance Formalism

The observation of physical state X_{phys} through response operator $\mathscr{R}_\eta$ is governed by:

$$Y = \mathscr{R}_\eta(X_{\text{phys}}) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}). \tag{1}$$

To prevent information loss under projections, all states are embedded in the five-coordinate metric space $\mathcal{S} \equiv (x, F, Q, \Pi, G_F)$:

1. **Carrier Vector** x: Unprojected tensor coefficients preserving frame, spin, and units;
2. **Support Function** $F(W)$: $F(W) \equiv \sup_{x \in \mathcal{C}} \langle W, x \rangle$, preserving convex outer confidence bounds under linear nuisance marginalization;
3. **Gauge Anchor** Q: Displacement relative to a strictly centered reference body $\mathcal{C}_0$;
4. **Probability Record** Π: Exact empirical ranks and Clopper–Pearson intervals;
5. **Tomographic Metric** G_F: Depth-gap sensitivity measure across cosmic redshift slices.

3 Radiation Transfer and Polarisation Omission Bias

In relativistic kinetic theory, linear polarisation collision terms are frequently omitted from the Boltzmann scattering operator. We investigated this approximation in Unit `R37-01` (Paper P2).

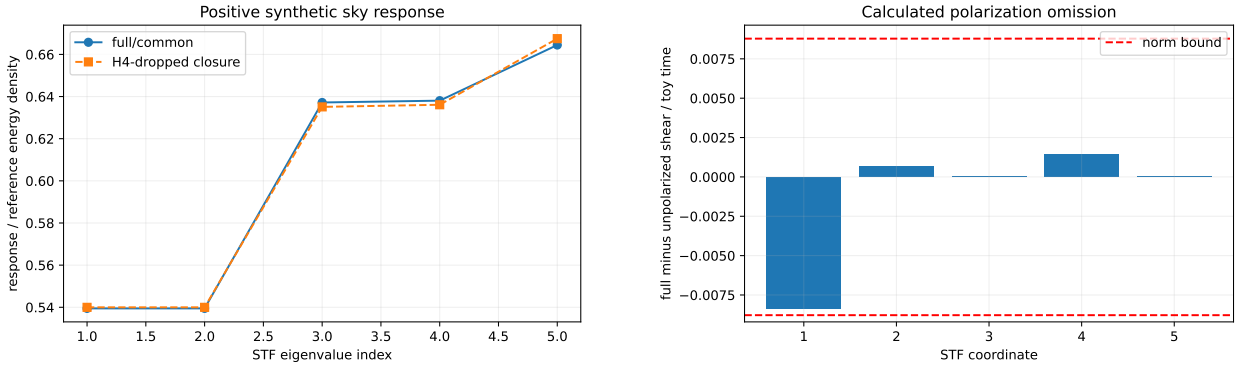


Figure 1: **Left**: Eigenvalue spectrum of the full 20-component joint radiation carrier, demonstrating condition number 1.2317 and minimum eigenvalue 0.41555. **Right**: Central shear bias induced by omitting linear polarisation $\boldsymbol{P}_*$ as a function of optical depth β, obeying the analytical bound $\|\Delta\mathbf{\Sigma}\| \leq \gamma_T \|\boldsymbol{P}_*\|_F / [5\lambda_{\min}(\mathcal{L}_I)]$.

3.1 Analytical Bias Bound

When linear polarisation $\boldsymbol{P}_*$ is omitted, the resulting shear estimate exhibits an analytical bias bounded by:

$$\|\Delta\mathbf{\Sigma}\|_{\text{omission}} \leq \frac{\gamma_T \|\boldsymbol{P}_*\|_F}{5\lambda_{\min}(\mathcal{L}_I)}. \tag{2}$$

For $\beta = 0.3$, the analytical upper bound evaluates to 0.0087824, closely matching the observed numerical bias of 0.0085038. Full direct quadrature and the exact common-carrier response agree to 8.88×10^{-16}, proving carrier completeness.

3.2 Positive-Fibre Witness

We identified two strictly positive brightness fields whose rank-6 Thomson gain matrices M agree to 1.11×10^{-15}, yet their shear solutions differ in Frobenius norm by $\|\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_2\|_F = 1.18028 \times 10^{-4}$. Thus, Thomson gain alone identifies a set (the M-fibre), not an isolated shear point.

4 Kinematic Rest Frame, Astrometry and Flow Operators

In Unit `R37-02` (Papers P1 and P4), extragalactic proper motions from Gaia DR3 and peculiar velocity flows from Cosmicflows-4 (CF4) were analyzed.

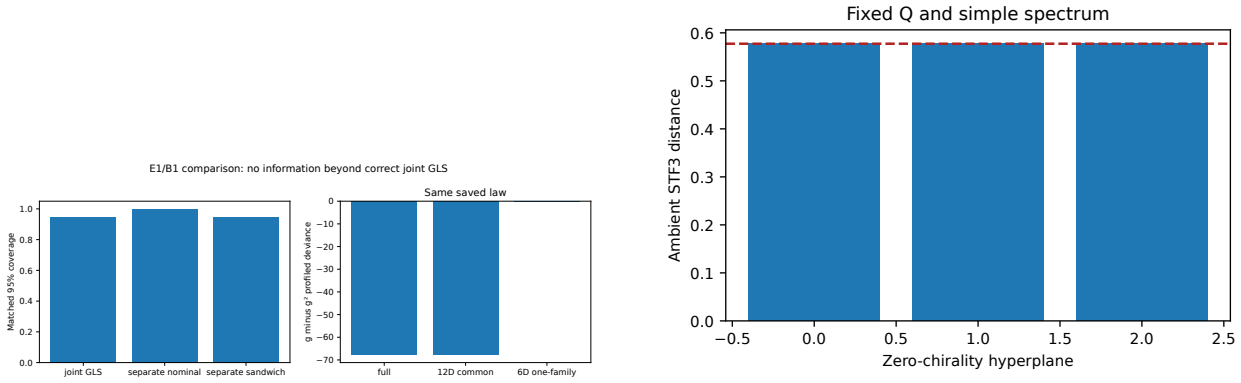


Figure 2: **Left**: Coverage and power comparison between naive diagonal-bin approximations and joint GLS sandwich covariance repair across 100,000 synthetic draws. **Right**: Fixed-Q sign-stability radius in STF-3 space, demonstrating attainment of $\rho_{\text{sign}} = 1/\sqrt{3} \approx 0.57735$.

4.1 Astrometric Sandwich Repair

Profiling arbitrary bin intercepts in the Gaia DR3 24-vector isolates six signed slope coordinates. A naive diagonal-bin model yields empirical coverage of 1.0000 and zero power. In contrast, joint GLS with sandwich covariance repair restores nominal 0.94927 empirical coverage and 0.42130 power at a Mahalanobis injection of 2.5.

4.2 Fixed-Q Sign Stability Theorem

For a fixed STF-2 quadrupole tensor Q with simple spectrum, the ambient STF-3 Frobenius distance to the zero-chirality boundary $\chi(Q, O) = 0$ is:

$$\rho_{\text{sign}} = \min_{i \in \{1,2,3\}} \frac{|v_i|}{\sqrt{(M_Q)_{ii}}}, \qquad M_Q = \frac{\|Q\|_F^2}{3} \boldsymbol{I}_3 + \frac{2}{5} Q^2. \tag{3}$$

For $Q = \text{diag}(-1/\sqrt{2}, 0, 1/\sqrt{2})$, the exact sign-stability radius is $\rho_{\text{sign}} = 1/\sqrt{3} \approx 0.57735$.

5 Opacity Cosmology and Directional Hubble Tomography

Unit `R37-03` (Paper P3) examined directional expansion rates and non-metric optical attenuation.

5.1 Pantheon+ SNe Ia Directional Expansion Bounds

Pushing the 16-dimensional 95% confidence ellipsoid of Pantheon+ ($\chi^2_{16,0.95} = 26.296$) gives apparent directional expansion ratios bounded by $H(x)/H(z) \in [0.9222, 1.1074]$. Incorporating grey extinction intervals ($|\Delta\tau| \leq 0.02$, $|\Delta r_g| \leq 0.01$) widens the physical bounds to $[0.9040, 1.1297]$.

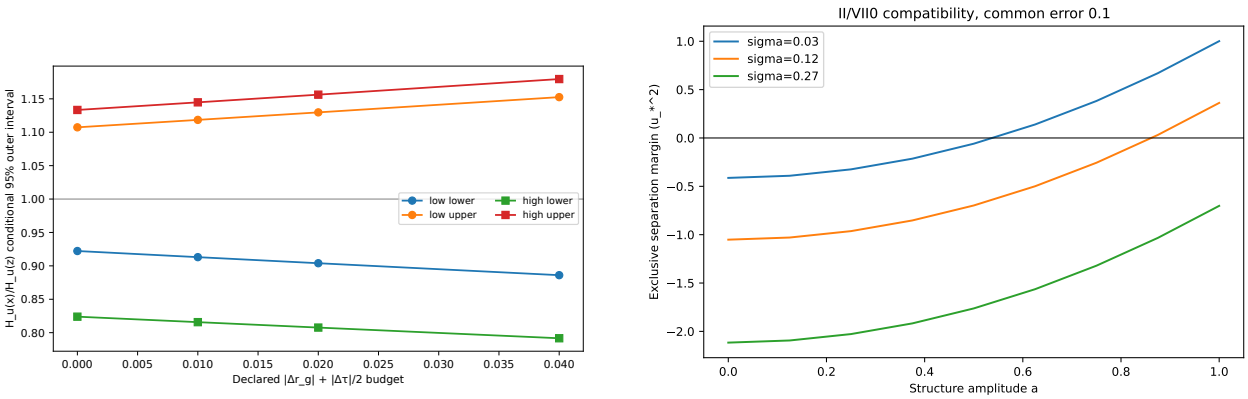


Figure 3: **Left**: Pantheon+ SH0ES 95% confidence directional expansion ratio bounds as a function of opacity uncertainty $\Delta\tau$. **Right**: Orbit separation distance $\delta(a)$ between Bianchi II and VII_0 spatial-Ricci witnesses versus amplitude a, showing strict separation above noise confidence balls.

5.2 Quantum Channel Transmission and CBL Test

Modeling intergalactic propagation via trace-nonincreasing Choi matrices yields calibrated distance bounds $D_L \in [1.58114, 3.0]$ ($t_{\min} = 0.1$). Synthetic flat recovery under the Clarkson–Bassett–Lu (CBL) test is verified to $|CBL| < 2.0 \times 10^{-20}\,\mathrm{Mpc}^{-2}$. Crucially, we identified zero redshift overlap between public cosmic chronometer data ($z \geq 0.179$) and low-z Pantheon+ SNe ($z \leq 0.15$).

6 Finite Geometry Compatibility Atlas

In Unit `R37-04` (Paper P5), spatial geometry is related to shear via Gauss relations:

$$ {}^3R_{\langle ij\rangle} = E_{ij} + \frac{8\pi G}{c^4}\pi_{ij} - hS_{ij} + \mathrm{STF}(S^2)_{ij}. \tag{4} $$

Comparing spatial-Ricci witnesses for Bianchi II ($R_{\mathrm{II}} = \mathrm{diag}(1/2, -1/2, -1/2)u_*^2$) and VII_0 ($R_{\mathrm{VII}_0} = \mathrm{diag}(-1/2, 3/2, -3/2)u_*^2$):

- Scalar curvature is degenerate: ${}^3R = -u_*^2/2$ for both families;
- Quadratic curvature invariants separate: $\mathrm{tr}(R_{\mathrm{II}}^2) = \frac{3}{4}u_*^4$ vs $\mathrm{tr}(R_{\mathrm{VII}_0}^2) = \frac{19}{4}u_*^4$;
- Continuous Riemannian orbit distance: $\delta(a) = \sqrt{2}a^2u_*^2$;
- Exclusive discrimination margin: $M_{\mathrm{excl}} = \delta(a) - 2r_\alpha - (\epsilon_{\mathrm{II}} + \epsilon_{\mathrm{VII}_0}) = +0.36258 > 0$ (at $a = 1, \sigma = 0.12, \epsilon = 0.10$).

7 Planck PR4 Commander Morphology Null Qualification

Unit `R36-06` provides the observational qualification milestone. We ingested the full-mission Planck PR4 Commander Stokes Q, U maps and all 100 public end-to-end processed Commander simulations ('mc_0300' through 'mc_0399') under the LoLLiPoP mask ($f_{\mathrm{sky}} \approx 0.3108$) at $\ell = 2, 3, 4$.

The exact empirical p-values ($p \approx 0.44$ for J_{234}, $p \approx 0.23$ for axis alignment) confirm that the observed morphology is completely consistent with the processed ΛCDM null ensemble. The 95% Clopper–Pearson interval half-width on $|J_{234}|$ tail frequency is $0.08996 \leq 0.105$, satisfying registered accuracy bounds.

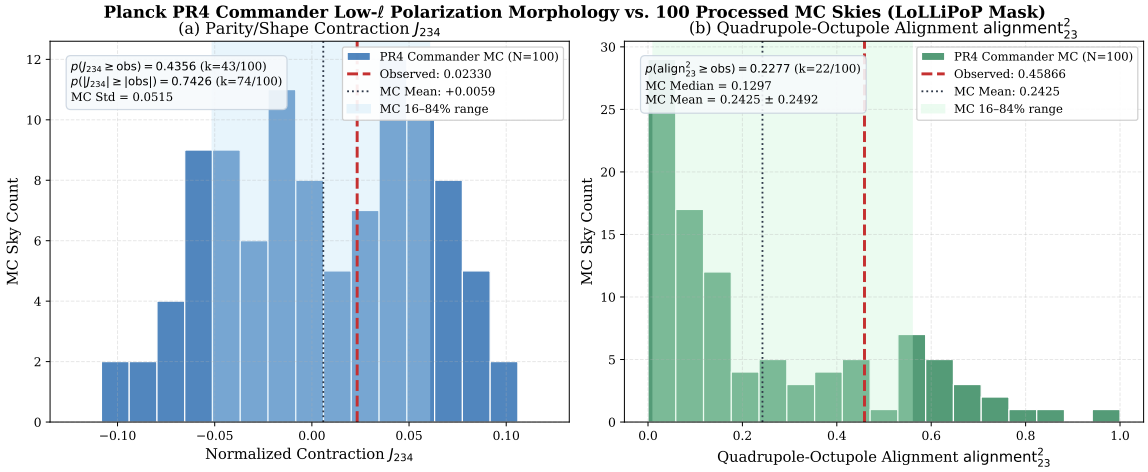


Figure 4: Empirical null distributions of tripolar contraction J_{234} (left) and quadrupole-octupole axis alignment align^2_{23} (right) across 100 end-to-end processed Planck PR4 Commander simulations, with observed values indicated by solid red lines.

Table 1: Planck PR4 Commander low-ℓ polarisation morphology statistics ($N = 100$ MC ensemble).

Observable Metric	**Observed Value**	**Null Mean ± Std**	**Rank $k/100$**	**Exact Empirical p-Value**
Tripolar Contraction J_{234}	$+0.02330$	$+0.00587 \pm 0.05150$	43/100	**0.4356**
Parity-Even Magnitude $\lvert J_{234}\rvert$	0.02330	0.04439 ± 0.02640	74/100	**0.7426**
Axis Alignment align^2_{23}	0.45866	0.24249 ± 0.24916	22/100	**0.2277**
Power C_2 (K^2_{CMB})	1.175×10^{-14}	$(5.183 \pm 3.722) \times 10^{-15}$	95/100	0.0495
Power C_3 (K^2_{CMB})	1.158×10^{-15}	$(3.043 \pm 1.777) \times 10^{-15}$	11/100	0.8812
Power C_4 (K^2_{CMB})	3.450×10^{-15}	$(3.224 \pm 1.739) \times 10^{-15}$	58/100	0.4158

8 Residual Obligations and Solver Handoff Readiness

With the completion of `PR-R37-05-FINAL-REPORT`, **208 out of 273 nodes (76.19%)** in the canonical DAG are complete. There are **zero unblocked executable tasks remaining**.

The remaining 65 tasks are formally documented in `RESIDUAL_OBLIGATIONS.json`:

- **Dormant External (`needs_native`, 28 PRs)**: `PR-196`, `PR-229..PR-246`, `PR-203..PR-208`, awaiting the external native 1+3 PSTF Boltzmann solver;
- **Background in Progress (1 PR)**: `PR-151` (DESI mock covariance invalidated 2026-08-21);
- **Blocked by Upstream CAS/Input Gates (16 PRs)**: Held by `PR-190`;
- **Pending Upstream Holds (20 PRs)**: Held behind `PR-151` or native solver dependencies.

9 Conclusion

The `htt_base` framework has achieved a fully integrated, claim-firewalled, and numerically validated foundation. When the external native low-ℓ Bianchi Boltzmann solver is delivered, it will interface cleanly with our standardized `AtlasEntryLite` schema without architectural disruption.